

The total antioxidant content of more than 3100 foods, beverages, spices, herbs and supplements used worldwide

Background A plant-based diet protects against chronic oxidative stress-related diseases. Dietary plants contain variable chemical families and amounts of antioxidants. It has been hypothesized that plant antioxidants may contribute to the beneficial health effects of dietary plants. Our objective was to develop a comprehensive food database consisting of the total antioxidant content of typical foods as well as other dietary items such as traditional medicine plants, herbs and spices and dietary supplements. This database is intended for use in a wide range of nutritional research, from in vitro and cell and animal studies, to clinical trials and nutritional epidemiological studies. **Methods** We procured samples from countries worldwide and assayed the samples for their total antioxidant content using a modified version of the FRAP assay. Results and sample information (such as country of origin, product and/or brand name) were registered for each individual food sample and constitute the Antioxidant Food Table. **Results** The results demonstrate that there are several thousand-fold differences in antioxidant content of foods. Spices, herbs and supplements include the most antioxidant rich products in our study, some exceptionally high. Berries, fruits, nuts, chocolate, vegetables and products thereof constitute common foods and beverages with high antioxidant values. **Conclusions** This database is to our best knowledge the most comprehensive Antioxidant Food Database published and it shows that plant-based foods introduce significantly more antioxidants into human diet than non-plant foods. Because of the large variations observed between otherwise comparable food samples the study emphasizes the importance of using a comprehensive database combined with a detailed system for food registration in clinical and epidemiological studies. The present antioxidant database is therefore an essential research tool to further elucidate the potential health effects of phytochemical antioxidants in diet.